



Development Setup Guide for Salon Booking System

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1. Recommended Folder Structure

1.1 Backend Folder Structure

salon-appointment-system/

```
|
|
| └─ backend/                # Node.js + Express + TypeScript
|
|   └─ src/
|
|     └─ config/
|
|       └─ database.ts        # MongoDB connection
|
|       └─ environment.ts     # Environment variables
|
|       └─ constants.ts       # App constants
|
|       └─ cors.ts            # CORS configuration
|
|       |
|
|     └─ models/              # Mongoose schemas
|
|       └─ Customer.ts
|
|       └─ Salon.ts
|
|       └─ Employee.ts
|
|       └─ Appointment.ts
|
|       └─ Service.ts
|
|       └─ GlobalService.ts
|
|       └─ ServiceCategory.ts
|
|       └─ EmployeeSchedule.ts
|
|       └─ TimeSlot.ts
|
|       └─ Feedback.ts
|
|       └─ Admin.ts
|
|       └─ Waitlist.ts
```

```

| | | └─ Blacklist.ts
| | | └─ FamilyMember.ts
| | | └─ CustomerAnalytics.ts
| | | └─ SubscriptionModule.ts
| | | └─ Payment.ts
| | | └─ Discount.ts
| | | └─ Promotion.ts
| | | └─ LoyaltyPoints.ts
| | | └─ Notification.ts
| | | └─ AuditTrail.ts
| | | └─ LoginAttempt.ts
| | | └─ ApiAccess.ts
| | | └─ index.ts
| | |
| | └─ controllers/           # Route handlers
| |   └─ auth/
| |     └─ customerAuth.controller.ts
| |     └─ salonOwnerAuth.controller.ts
| |     └─ adminAuth.controller.ts
| |   └─ customer/
| |     └─ profile.controller.ts
| |     └─ booking.controller.ts
| |     └─ appointment.controller.ts
| |     └─ feedback.controller.ts
| |     └─ family.controller.ts

```

```

| | | | └─ preferences.controller.ts
| | | | └─ salon/
| | | |   └─ profile.controller.ts
| | | |   └─ service.controller.ts
| | | |   └─ appointment.controller.ts
| | | |   └─ employee.controller.ts
| | | |   └─ schedule.controller.ts
| | | |   └─ analytics.controller.ts
| | | |   └─ feedback.controller.ts
| | | | └─ admin/
| | | |   └─ dashboard.controller.ts
| | | |   └─ salon.controller.ts
| | | |   └─ customer.controller.ts
| | | |   └─ analytics.controller.ts
| | | |   └─ moderation.controller.ts
| | | |   └─ globalService.controller.ts
| | | |   └─ reports.controller.ts
| | | | └─ common/
| | | |   └─ search.controller.ts
| | | |   └─ qr.controller.ts
| | | |   └─ upload.controller.ts
| | | | └─ payment/
| | | |   └─ payment.controller.ts
| | | |   └─ promotion.controller.ts
| | |

```

```

| | ├── routes/           # API routes
| | |   ├── v1/
| | |   |   ├── auth.routes.ts
| | |   |   ├── customer.routes.ts
| | |   |   ├── salon.routes.ts
| | |   |   ├── admin.routes.ts
| | |   |   ├── appointment.routes.ts
| | |   |   ├── feedback.routes.ts
| | |   |   ├── payment.routes.ts
| | |   |   ├── promotion.routes.ts
| | |   |   ├── loyalty.routes.ts
| | |   |   ├── notification.routes.ts
| | |   |   ├── search.routes.ts
| | |   |   ├── qr.routes.ts
| | |   |   └── index.ts
| | |   └── index.ts
| | |
| | ├── middlewares/      # Custom middlewares
| | |   ├── auth.middleware.ts    # JWT verification
| | |   ├── validation.middleware.ts # Request validation
| | |   ├── errorHandler.middleware.ts # Error handling
| | |   ├── roleCheck.middleware.ts # Role-based access
| | |   ├── rateLimit.middleware.ts # Rate limiting
| | |   ├── upload.middleware.ts   # File upload (multer)
| | |   └── logger.middleware.ts   # Request logging

```

```

| | | └─ sanitize.middleware.ts    # Input sanitization
| | |
| | | └─ services/                # Business logic
| | |   └─ auth.service.ts
| | |   └─ booking.service.ts
| | |   └─ notification.service.ts
| | |   └─ payment.service.ts
| | |   └─ sms.service.ts         # Twilio integration
| | |   └─ email.service.ts       # Nodemailer
| | |   └─ analytics.service.ts
| | |   └─ qr.service.ts          # QR code generation
| | |   └─ search.service.ts      # Advanced search
| | |   └─ file.service.ts        # File handling
| | |
| | | └─ utils/                  # Helper functions
| | |   └─ encryption.ts          # bcrypt
| | |   └─ jwt.ts                 # JWT utilities
| | |   └─ otp.ts                 # OTP generation
| | |   └─ dateTime.ts            # Date helpers
| | |   └─ validation.ts          # Validators
| | |   └─ logger.ts              # Winston logger
| | |   └─ response.ts            # Response formatter
| | |   └─ gps.ts                 # GPS calculations
| | |
| | | └─ types/                  # TypeScript types

```

```

| | | ├── models.d.ts      # Model interfaces
| | | ├── request.d.ts     # Request types
| | | ├── response.d.ts    # Response types
| | | ├── express.d.ts     # Express extensions
| | | └── index.ts
| | |
| | | ├── validators/      # Request validators (Zod)
| | | |   ├── auth.validator.ts
| | | |   ├── booking.validator.ts
| | | |   ├── salon.validator.ts
| | | |   ├── customer.validator.ts
| | | |   └── appointment.validator.ts
| | |
| | | ├── jobs/           # Background jobs (node-cron)
| | | |   ├── reminderCron.ts    # Appointment reminders
| | | |   ├── offerExpiry.ts     # Expire old offers
| | | |   ├── dataCleanup.ts     # Clean old records
| | | |   ├── notificationQueue.ts # Process notifications
| | | |   └── index.ts
| | |
| | | ├── app.ts          # Express app setup
| | | └── server.ts        # Server entry point
| |
| ├── tests/             # Test files
| └── unit/

```



```

| | | └─ services/
| | | └─ controllers/
| | | └─ utils/
| | └─ integration/
| | | └─ api/
| | └─ e2e/
| | └─ flows/
| |
| └─ uploads/          # File uploads (gitignored)
| | └─ profiles/
| | └─ salons/
| | └─ qr-codes/
| |
| └─ .env.example
| └─ .env
| └─ .gitignore
| └─ package.json
| └─ tsconfig.json
| └─ nodemon.json
| └─ README.md
|
└─ frontend-web/      # React.js Web Applications
| |
| └─ customer-web/    # Customer Web Portal
| | └─ src/

```

```

| | | |└─ assets/
| | | |└─ images/
| | | |└─ icons/
| | | |└─ fonts/
| | | |└─ components/
| | | |└─ common/          # Shared components
| | | | |└─ Button/
| | | | |└─ Input/
| | | | |└─ Card/
| | | | |└─ Modal/
| | | | |└─ Loader/
| | | | |└─ Navbar/
| | | | |└─ auth/
| | | | |└─ LoginForm.tsx
| | | | |└─ RegisterForm.tsx
| | | | |└─ OTPVerification.tsx
| | | | |└─ booking/
| | | | |└─ SalonList.tsx
| | | | |└─ ServiceSelector.tsx
| | | | |└─ TimeSlotPicker.tsx
| | | | |└─ BookingConfirmation.tsx
| | | | |└─ appointments/
| | | | |└─ AppointmentCard.tsx
| | | | |└─ AppointmentDetails.tsx
| | | | |└─ profile/

```

```

| | | | | ├── ProfileInfo.tsx
| | | | | └── FamilyMembers.tsx
| | | | └── feedback/
| | | |     └── FeedbackForm.tsx
| | | ├── pages/
| | | | ├── auth/
| | | | | ├── Login.tsx
| | | | | ├── Register.tsx
| | | | | └── ForgotPassword.tsx
| | | | ├── home/
| | | | | └── Home.tsx
| | | | ├── salon/
| | | | | ├── SalonDiscovery.tsx
| | | | | └── SalonDetails.tsx
| | | | ├── booking/
| | | | | ├── BookingFlow.tsx
| | | | | └── BookingSuccess.tsx
| | | | ├── appointments/
| | | | | ├── MyAppointments.tsx
| | | | | └── AppointmentHistory.tsx
| | | | ├── feedback/
| | | | | └── LeaveFeedback.tsx
| | | | └── profile/
| | | |     └── Profile.tsx
| | | ├── hooks/           # Custom React hooks

```

```

| | | | ├── useAuth.ts
| | | | ├── useBooking.ts
| | | | ├── useAppointments.ts
| | | | └── useLanguage.ts
| | | ├── context/          # React Context
| | | | ├── AuthContext.tsx
| | | | ├── LanguageContext.tsx
| | | | └── ThemeContext.tsx
| | | ├── services/        # API calls
| | | | ├── api.ts          # Axios instance
| | | | ├── authService.ts
| | | | ├── bookingService.ts
| | | | ├── appointmentService.ts
| | | | └── salonService.ts
| | | ├── utils/
| | | | ├── dateFormatter.ts
| | | | ├── validation.ts
| | | | └── helpers.ts
| | | ├── types/
| | | | ├── api.types.ts
| | | | ├── customer.types.ts
| | | | └── appointment.types.ts
| | | ├── constants/
| | | | ├── api.constants.ts
| | | | └── routes.constants.ts

```

```

| | | └─ styles/
| | | | └─ globals.css
| | | └─ i18n/          # Internationalization
| | | | └─ en.json
| | | | └─ si.json
| | | | └─ i18n.ts
| | | └─ App.tsx
| | | └─ main.tsx
| | └─ public/
| | | └─ index.html
| | | └─ favicon.ico
| | └─ package.json
| | └─ vite.config.ts
| | └─ tsconfig.json
| | └─ tailwind.config.js
| | └─ README.md
| |
| └─ salon-owner-web/    # Salon Owner Web Portal
| | └─ src/
| | | └─ assets/
| | | └─ components/
| | | | └─ common/
| | | | └─ dashboard/
| | | | | └─ StatsCard.tsx
| | | | | └─ RevenueChart.tsx

```

```

| | | | | └─ appointments/
| | | | |   └─ AppointmentCalendar.tsx
| | | | |     └─ AppointmentList.tsx
| | | | | └─ services/
| | | | |   └─ ServiceList.tsx
| | | | |     └─ ServiceForm.tsx
| | | | | └─ employees/
| | | | |   └─ EmployeeList.tsx
| | | | |   └─ EmployeeForm.tsx
| | | | |     └─ ScheduleManager.tsx
| | | | | └─ analytics/
| | | | |   └─ AnalyticsCharts.tsx
| | | | | └─ pages/
| | | | |   └─ auth/
| | | | |   └─ dashboard/
| | | | |     └─ Dashboard.tsx
| | | | | └─ appointments/
| | | | |   └─ Appointments.tsx
| | | | | └─ services/
| | | | |   └─ Services.tsx
| | | | | └─ employees/
| | | | |   └─ Employees.tsx
| | | | | └─ analytics/
| | | | |   └─ Analytics.tsx
| | | | | └─ feedback/

```

```

| | | | | └─ Feedback.tsx
| | | | | └─ settings/
| | | | |   └─ Settings.tsx
| | | └─ hooks/
| | | └─ services/
| | | └─ utils/
| | | └─ types/
| | | └─ constants/
| | | └─ i18n/
| | | └─ App.tsx
| | | └─ main.tsx
| | └─ public/
| | └─ package.json
| | └─ vite.config.ts
| | └─ tsconfig.json
| | └─ tailwind.config.js
| |
| └─ admin-web/          # Admin Web Portal
|   └─ src/
|     └─ assets/
|     └─ components/
|       └─ common/
|       └─ dashboard/
|         └─ DashboardStats.tsx
|         └─ salons/

```

- | | | | └─ SalonList.tsx
- | | | | └─ SalonDetails.tsx
- | | | | └─ ApprovalQueue.tsx
- | | | └─ customers/
- | | | | └─ CustomerList.tsx
- | | | └─ analytics/
- | | | | └─ PlatformAnalytics.tsx
- | | | └─ moderation/
- | | | | └─ FeedbackModeration.tsx
- | | └─ pages/
- | | | └─ auth/
- | | | | └─ AdminLogin.tsx
- | | | └─ dashboard/
- | | | | └─ Dashboard.tsx
- | | | └─ salons/
- | | | | └─ SalonManagement.tsx
- | | | | └─ PendingApprovals.tsx
- | | | └─ customers/
- | | | | └─ CustomerManagement.tsx
- | | | └─ feedback/
- | | | | └─ FeedbackModeration.tsx
- | | | └─ analytics/
- | | | | └─ Analytics.tsx
- | | | └─ reports/
- | | | | └─ Reports.tsx


```
| |   ├── hooks/  
| |   ├── services/  
| |   ├── utils/  
| |   ├── types/  
| |   ├── constants/  
| |   └── App.tsx  
| |  
|     └─ main.tsx  
  
├── public/  
├── package.json  
├── vite.config.ts  
└─ tsconfig.json  
  
└─ mobile-app/                                # Flutter Mobile Applications  
    │  
    │  
    ├── customer_mobile/                      # Customer Mobile App  
    │  
    │   ├── android/  
    │   ├── ios/  
    │   ├── lib/  
    │   │   ├── main.dart  
    │   │   ├── core/  
    │   │   │   ├── constants/  
    │   │   │   │   ├── api_constants.dart  
    │   │   │   │   └─ app_constants.dart  
    │   │   └── config/  
    │   │       └─ app_config.dart
```

```

| | | | └─ utils/
| | | | | └─ date_formatter.dart
| | | | | └─ validators.dart
| | | | └─ theme/
| | | | └─ app_theme.dart
| | | └─ data/
| | | | └─ models/
| | | | | └─ customer_model.dart
| | | | | └─ salon_model.dart
| | | | | └─ appointment_model.dart
| | | | | └─ service_model.dart
| | | | └─ repositories/
| | | | | └─ auth_repository.dart
| | | | | └─ booking_repository.dart
| | | | | └─ salon_repository.dart
| | | | └─ api/
| | | | | └─ api_client.dart
| | | | | └─ endpoints.dart
| | | └─ presentation/
| | | | └─ screens/
| | | | | └─ auth/
| | | | | | └─ login_screen.dart
| | | | | | └─ register_screen.dart
| | | | | | └─ otp_verification_screen.dart
| | | | | └─ home/

```

```

| | | | | | └─ home_screen.dart
| | | | | | └─ salon/
| | | | | |   └─ salon_list_screen.dart
| | | | | |   └─ salon_details_screen.dart
| | | | | | └─ booking/
| | | | | |   └─ booking_flow_screen.dart
| | | | | |   └─ booking_success_screen.dart
| | | | | | └─ appointments/
| | | | | |   └─ my_appointments_screen.dart
| | | | | |   └─ appointment_details_screen.dart
| | | | | | └─ profile/
| | | | | |   └─ profile_screen.dart
| | | | | | └─ feedback/
| | | | | |   └─ feedback_screen.dart
| | | | | | └─ widgets/
| | | | | | └─ common/
| | | | | |   └─ custom_button.dart
| | | | | |   └─ custom_input.dart
| | | | | |   └─ loading_indicator.dart
| | | | | | └─ salon/
| | | | | |   └─ salon_card.dart
| | | | | |   └─ appointment/
| | | | | |     └─ appointment_card.dart
| | | | └─ providers/      # State Management (Provider/Riverpod)
| | | └─ └─ auth_provider.dart

```

```

| | | |   ├── booking_provider.dart
| | | |   └── language_provider.dart
| | |   ├── routes/
| | |   └── app_routes.dart
| |   └── localization/
| |     ├── en.json
| |     └── si.json
| |   ├── assets/
| |   ├── images/
| |   └── icons/
| |   ├── pubspec.yaml
| |   └── README.md
| |
| |   └── salon_owner_mobile/      # Salon Owner Mobile App
| |     ├── android/
| |     ├── ios/
| |     ├── lib/
| |     ├── main.dart
| |     ├── core/
| |     ├── constants/
| |     ├── config/
| |     ├── utils/
| |     ├── theme/
| |     ├── data/
| |     └── models/

```

```
| | | | └─ salon_model.dart
| | | | └─ employee_model.dart
| | | | └─ service_model.dart
| | | |   └─ appointment_model.dart
| | | └─ repositories/
| | |   └─ auth_repository.dart
| | |   └─ salon_repository.dart
| | |   └─ appointment_repository.dart
| | |     └─ employee_repository.dart
| | └─ api/
| |   └─ api_client.dart
| └─ presentation/
|   └─ screens/
|     └─ auth/
|       └─ login_screen.dart
|       └─ dashboard/
|         └─ dashboard_screen.dart
|         └─ appointments/
|           └─ appointments_screen.dart
|           └─ appointment_calendar_screen.dart
|           └─ services/
|             └─ services_screen.dart
|             └─ employees/
|               └─ employees_screen.dart
|               └─ schedule_screen.dart
```



```

|   ├── api/
|   |   ├── authentication.md
|   |   ├── customer-endpoints.md
|   |   ├── salon-endpoints.md
|   |   └── admin-endpoints.md
|   ├── architecture/
|   |   ├── system-design.md
|   |   ├── database-schema.md
|   |   └── api-flow-diagrams.md
|   ├── deployment/
|   |   ├── backend-deployment.md
|   |   ├── web-deployment.md
|   |   └── mobile-deployment.md
|   └── development/
|       ├── setup-guide.md
|       ├── coding-standards.md
|       └── testing-guide.md
|
|── scripts/                # Utility scripts
|   ├── seed-data.js        # Seed database
|   ├── migrate.js          # Run migrations
|   └── generate-qr.js       # Generate QR codes
|
|── .gitignore
|── .env.example

```

└─ docker-compose.yml # Docker setup for backend
└─ README.md
└─ package.json # Root package.json (if using monorepo tools)

2. Development Phases Based on BRD

Stage 1: Core MVP (Weeks 1-8)

Focus on Appointment Management & Feedback

Backend Priority:

1. Authentication System

- Customer registration/login
- Salon owner registration/login
- Admin login
- OTP verification

2. Salon Management

- Salon profile CRUD
- Service management (from global list)
- Employee management
- Time slot management

3. Appointment System

- Self booking flow
- Family booking flow
- View appointments
- Cancel/reschedule appointments
- Appointment reminders

4. Feedback System

- Post-service reviews
- Rating system
- Feedback moderation

Frontend Priority:

1. Customer App

- Language selection
- Authentication screens
- Salon listing & search
- Booking flow (self & family)
- My appointments
- Feedback submission

2. Salon Owner App

- Dashboard
- Service management
- Appointment calendar
- Staff management
- Feedback view

3. Admin App

- Dashboard
- Salon approval system
- User management
- Feedback moderation

Stage 2: Payment & Promotions (Weeks 9-12)

1. Payment gateway integration
2. Promotion management system
3. Financial analytics

Stage 3: Loyalty System (Weeks 13-15)

1. Points earning mechanism
2. Points redemption
3. Loyalty tracking

Stage 4: Employee Management Enhancement (Weeks 16-18)

1. Advanced scheduling
2. Performance tracking
3. Multi-salon employee support

Technology Stack Setup

Web Frontend

```
{  
  "dependencies": {  
    "react": "^18.x",  
    "react-router-dom": "^6.x",  
    "axios": "^1.x",  
    "react-hook-form": "^7.x",  
    "zod": "^3.x",  
    "zustand": "^4.x",  
    "react-query": "^3.x",  
    "i18next": "^23.x",  
    "react-i18next": "^13.x",  
    "date-fns": "^2.x",  
    "tailwindcss": "^3.x"  
  }  
}
```

Mobile App (Flutter)

pubspec.yaml

dependencies:

flutter:

 sdk: flutter

State Management

provider: ^6.1.0 # or riverpod: ^2.4.0

HTTP Client

dio: ^5.3.0

http: ^1.1.0

Local Storage

shared_preferences: ^2.2.0

hive: ^2.2.3

Navigation

go_router: ^12.0.0

Internationalization

flutter_localizations:

 sdk: flutter

intl: ^0.18.0

UI/UX

flutter_svg: ^2.0.0

cached_network_image: ^3.3.0

Form Validation

flutter_form_builder: ^9.1.0

Date/Time

intl: ^0.18.0

QR Code

qr_flutter: ^4.1.0

qr_code_scanner: ^1.0.1

Push Notifications

firebase_messaging: ^14.7.0

Maps/Location

google_maps_flutter: ^2.5.0

geolocator: ^10.1.0

Backend

```
{  
  "dependencies": {  
    "express": "^4.x",  
    "mongoose": "^8.x",  
    "bcryptjs": "^2.x",  
    "jsonwebtoken": "^9.x",  
    "dotenv": "^16.x",  
    "cors": "^2.x",  
    "helmet": "^7.x",  
    "express-rate-limit": "^7.x",  
    "nodemailer": "^6.x",  
    "twilio": "^4.x",  
    "node-cron": "^3.x",  
    "zod": "^3.x"  
  }  
}
```

Getting Started Steps

1. Initialize Backend

```
mkdir salon-backend
```

```
cd salon-backend
```

```
npm init -y
```

```
npm install express mongoose typescript @types/node @types/express ts-node nodemon
```

```
npm run tsc --init
```

2. Initialize Frontend Web App

```
# Navigate to frontend-web directory
```

```
cd frontend-web
```

```
# Customer Web
```

```
npm create vite@latest customer-web -- --template react-ts
```

```
# Salon Owner Web
```

```
npm create vite@latest salon-owner-web -- --template react-ts
```

```
# Admin Web
```

```
npm create vite@latest admin-web -- --template react-ts
```


For Mobile Apps (Flutter)

Navigate to mobile-app directory

```
cd mobile-app
```

Customer Mobile

```
flutter create customer_mobile
```

```
cd customer_mobile
```

Configure as per architecture

Salon Owner Mobile

```
cd ..
```

```
flutter create salon_owner_mobile
```

```
cd salon_owner_mobile
```

Configure as per architecture

3. Setup MongoDB Atlas

- Create cluster
- Setup database user
- Whitelist IP
- Get connection string

4. Environment Variables

Backend .env

PORT=5000

MONGODB_URI=mongodb+srv://...

JWT_SECRET=your_secret_key

JWT_EXPIRE=7d

OTP_EXPIRY=10

TWILIO_ACCOUNT_SID=...

TWILIO_AUTH_TOKEN=...

SMTP_HOST=...

SMTP_PORT=...

Database Schema Considerations

Since you're using MongoDB (not relational), adjust your schema design:

Example: Appointment Model

```
interface IAppointment {  
  customerId: ObjectId;  
  salonId: ObjectId;  
  employeeId?: ObjectId;  
  services: Array<{  
    serviceId: ObjectId;  
    serviceName: string;  
    duration: number;  
    price: number;  
  }>;  
  appointmentDate: Date;  
  startTime: string;  
  endTime: string;  
  status: 'booked' | 'confirmed' | 'completed' | 'cancelled';  
  bookingType: 'single' | 'family';  
  familyMembers?: Array<{  
    name: string;  
    gender: string;  
    ageCategory: string;  
    services: ObjectId[];  
  }>;  
}
```

Next Immediate Steps

1. **Week 1:** Setup project structure + authentication
2. **Week 2:** Implement salon registration & service management
3. **Week 3-4:** Build booking system (self & family flows)
4. **Week 5:** Appointment management & notifications
5. **Week 6:** Feedback system
6. **Week 7-8:** Testing & refinement

API Endpoints Documentation

Base URL: <http://localhost:5000/api/v1>

Authentication Endpoints

Customer Authentication

- POST /auth/customer/register - Register new customer
- POST /auth/customer/login - Customer login
- POST /auth/customer/guest-login - Guest customer login (phone only)
- POST /auth/customer/verify-otp - Verify OTP
- POST /auth/customer/resend-otp - Resend OTP
- POST /auth/customer/forgot-password - Request password reset
- POST /auth/customer/reset-password - Reset password with token
- POST /auth/customer/logout - Logout customer
- GET /auth/customer/refresh-token - Refresh access token
- POST /auth/customer/social-login - Google/Facebook login

Salon Owner Authentication

- POST /auth/salon/register - Register new salon owner
- POST /auth/salon/login - Salon owner login
- POST /auth/salon/verify-otp - Verify OTP
- POST /auth/salon/resend-otp - Resend OTP
- POST /auth/salon/forgot-password - Request password reset
- POST /auth/salon/reset-password - Reset password
- POST /auth/salon/logout - Logout salon owner
- GET /auth/salon/refresh-token - Refresh access token
- PUT /auth/salon/change-password - Change password (first login)

Admin Authentication

POST /auth/admin/login - Admin login
POST /auth/admin/logout - Admin logout
POST /auth/admin/forgot-password - Request password reset
POST /auth/admin/reset-password - Reset password
GET /auth/admin/refresh-token - Refresh access token

Customer Endpoints

Profile Management

GET /customer/profile - Get customer profile
PUT /customer/profile - Update customer profile
DELETE /customer/profile - Delete customer account
GET /customer/preferences - Get saved preferences
PUT /customer/preferences - Update preferences

Family Members

GET /customer/family-members - Get all family members
POST /customer/family-members - Add family member
PUT /customer/family-members/:id - Update family member
DELETE /customer/family-members/:id - Delete family member

Customer Analytics

GET /customer/analytics - Get customer analytics (visits, spending, etc.)
GET /customer/booking-history - Get booking history with filters

Salon Discovery & Information

Salon Listing

- GET /salons - Get all salons (with filters)
- GET /salons/nearby - Get nearby salons (GPS-based)
- GET /salons/search - Search salons by city/district
- GET /salons/:id - Get salon details
- GET /salons/:id/services - Get salon services
- GET /salons/:id/employees - Get available employees
- GET /salons/:id/reviews - Get salon reviews
- GET /salons/:id/availability - Get salon availability calendar

Global Services

- GET /services/global - Get global service list
- GET /services/global/:id - Get specific global service
- GET /services/categories - Get service categories

Appointment Management (Customer Side)

Booking

- POST /appointments/check-availability - Check time slot availability
- POST /appointments/book - Book appointment (self/family)
- GET /appointments - Get customer appointments
- GET /appointments/:id - Get appointment details
- PUT /appointments/:id/reschedule - Reschedule appointment
- DELETE /appointments/:id/cancel - Cancel appointment
- POST /appointments/:id/confirm - Confirm appointment reminder

Walk-in

POST /appointments/walk-in/qr-scan - QR code walk-in check-in

POST /appointments/walk-in/manual - Manual walk-in booking

Waitlist

POST /appointments/waitlist/join - Join waitlist

DELETE /appointments/waitlist/:id - Leave waitlist

GET /appointments/waitlist - Get customer waitlist entries

Salon Owner Endpoints

Salon Profile Management

GET /salon/profile - Get salon profile

PUT /salon/profile - Update salon profile

PUT /salon/profile/business-info - Update business information

PUT /salon/profile/working-hours - Update working hours

POST /salon/profile/images - Upload salon images

DELETE /salon/profile/images/:id - Delete salon image

Service Management

GET /salon/services - Get salon services

POST /salon/services - Add new service (from global list)

PUT /salon/services/:id - Update service (price, duration)

DELETE /salon/services/:id - Delete service

PUT /salon/services/:id/offer - Add/update offer on service

DELETE /salon/services/:id/offer - Remove offer

Employee Management

GET /salon/employees - Get all employees
POST /salon/employees - Add new employee
PUT /salon/employees/:id - Update employee
DELETE /salon/employees/:id - Delete/terminate employee
PUT /salon/employees/:id/status - Update employee status

Employee Schedule

GET /salon/employees/:id/schedule - Get employee schedule
POST /salon/employees/:id/schedule - Create employee schedule
PUT /salon/employees/:id/schedule - Update employee schedule
DELETE /salon/employees/:id/schedule/:scheduleId - Delete schedule

Time Slot Management

GET /salon/time-slots - Get time slot configuration
PUT /salon/time-slots - Update time slot settings
POST /salon/time-slots/block - Block specific time slots
DELETE /salon/time-slots/block/:id - Unblock time slot

Appointment Management (Salon Side)

GET /salon/appointments - Get all appointments (with filters)
GET /salon/appointments/today - Get today's appointments
GET /salon/appointments/upcoming - Get upcoming appointments
GET /salon/appointments/:id - Get appointment details
PUT /salon/appointments/:id/status - Update appointment status
DELETE /salon/appointments/:id/cancel - Cancel appointment (salon side)
POST /salon/appointments/:id/reschedule-request - Send reschedule request

Dashboard & Analytics

- GET /salon/dashboard - Get dashboard overview
- GET /salon/analytics/revenue - Get revenue analytics
- GET /salon/analytics/appointments - Get appointment statistics
- GET /salon/analytics/customers - Get customer analytics
- GET /salon/analytics/services - Get service performance
- GET /salon/analytics/employees - Get employee performance
- POST /salon/reports/download - Download reports (CSV/PDF)

Feedback Management

- GET /salon/feedback - Get all feedback
- GET /salon/feedback/:id - Get specific feedback
- POST /salon/feedback/:id/reply - Reply to feedback

Feedback & Review Endpoints

Customer Feedback

- POST /feedback - Submit feedback
- GET /feedback/my-feedback - Get customer's feedback history
- PUT /feedback/:id - Update feedback
- DELETE /feedback/:id - Delete feedback

Public Reviews

- GET /reviews/salon/:salonId - Get salon reviews (public)
- GET /reviews/:id - Get specific review

Notification Endpoints

Customer Notifications

- GET /notifications - Get all notifications
- GET /notifications/unread - Get unread notifications
- PUT /notifications/:id/read - Mark as read
- PUT /notifications/read-all - Mark all as read
- DELETE /notifications/:id - Delete notification

Notification Preferences

- GET /notifications/preferences - Get notification settings
- PUT /notifications/preferences - Update notification settings

Admin Endpoints

Dashboard

- GET /admin/dashboard - Get admin dashboard overview
- GET /admin/analytics/overview - Get platform analytics
- GET /admin/analytics/revenue - Get revenue statistics
- GET /admin/analytics/users - Get user statistics

Salon Management

- GET /admin/salons - Get all salons (with filters)
- GET /admin/salons/pending - Get pending registrations
- GET /admin/salons/:id - Get salon details
- PUT /admin/salons/:id/approve - Approve salon registration
- PUT /admin/salons/:id/reject - Reject salon registration
- PUT /admin/salons/:id/suspend - Suspend salon
- PUT /admin/salons/:id/activate - Activate salon
- DELETE /admin/salons/:id - Delete salon permanently
- POST /admin/salons/register - Register salon on behalf of owner

Customer Management

- GET /admin/customers - Get all customers
- GET /admin/customers/:id - Get customer details
- PUT /admin/customers/:id/block - Block customer
- PUT /admin/customers/:id/unblock - Unblock customer
- GET /admin/customers/:id/history - Get customer booking history

Blacklist Management

- GET /admin/blacklist - Get blacklisted customers
- POST /admin/blacklist - Add to blacklist
- DELETE /admin/blacklist/:id - Remove from blacklist

Feedback Moderation

- GET /admin/feedback/pending - Get pending feedback
- GET /admin/feedback/flagged - Get flagged feedback
- PUT /admin/feedback/:id/approve - Approve feedback
- PUT /admin/feedback/:id/reject - Reject feedback
- DELETE /admin/feedback/:id - Delete feedback

Global Service Management

- GET /admin/global-services - Get global service list
- POST /admin/global-services - Add global service
- PUT /admin/global-services/:id - Update global service
- DELETE /admin/global-services/:id - Delete global service

Category Management

- GET /admin/categories - Get all categories
- POST /admin/categories - Add category
- PUT /admin/categories/:id - Update category
- DELETE /admin/categories/:id - Delete category

Audit Trail

GET /admin/audit-trail - Get audit logs

GET /admin/audit-trail/:id - Get specific log

Login Attempt Logs

GET /admin/login-attempts - Get login attempt logs

GET /admin/login-attempts/failed - Get failed login attempts

Subscription Module Management

GET /admin/subscriptions - Get all subscriptions

GET /admin/subscriptions/:salonId - Get salon subscriptions

POST /admin/subscriptions - Create subscription

PUT /admin/subscriptions/:id - Update subscription

DELETE /admin/subscriptions/:id - Delete subscription

Reports

GET /admin/reports/salons - Get salon reports

GET /admin/reports/revenue - Get revenue reports

GET /admin/reports/customers - Get customer reports

POST /admin/reports/generate - Generate custom report

POST /admin/reports/download - Download report

Payment Endpoints (Stage 2)

Customer Payments

POST /payments/initiate - Initiate payment

POST /payments/verify - Verify payment

GET /payments/history - Get payment history

GET /payments/:id - Get payment details

POST /payments/refund - Request refund

Salon Payment Management

- GET /salon/payments - Get payment history
- GET /salon/payments/pending - Get pending payments
- GET /salon/payments/statistics - Get payment statistics

Admin Payment Management

- GET /admin/payments - Get all payments
- GET /admin/payments/statistics - Get payment statistics
- POST /admin/payments/:id/refund - Process refund

Promotion Endpoints (Stage 2)

Customer Promotions

- GET /promotions/active - Get active promotions
- GET /promotions/:id - Get promotion details
- POST /promotions/:id/apply - Apply promotion to booking

Salon Promotions

- GET /salon/promotions - Get salon promotions
- POST /salon/promotions - Create promotion
- PUT /salon/promotions/:id - Update promotion
- DELETE /salon/promotions/:id - Delete promotion
- POST /salon/promotions/:id/send - Send targeted promotion
- GET /salon/promotions/:id/analytics - Get promotion analytics

Admin Promotions

- GET /admin/promotions - Get all promotions
- GET /admin/promotions/statistics - Get promotion statistics

Loyalty Points Endpoints (Stage 3)

Customer Loyalty

- GET /loyalty/points - Get loyalty points balance
- GET /loyalty/history - Get points history
- POST /loyalty/redeem - Redeem points
- GET /loyalty/rewards - Get available rewards

Salon Loyalty Management

- GET /salon/loyalty/customers - Get customer loyalty data
- PUT /salon/loyalty/settings - Update loyalty settings
- POST /salon/loyalty/award - Award bonus points

Admin Loyalty

- GET /admin/loyalty/overview - Get loyalty program overview
- PUT /admin/loyalty/settings - Update global loyalty settings

Employee Management Advanced (Stage 4)

Multi-Salon Employee

- GET /employees/:id/salons - Get employee's salons
- POST /employees/:id/assign-salon - Assign to another salon
- DELETE /employees/:id/remove-salon - Remove from salon

Performance Tracking

- GET /salon/employees/:id/performance - Get employee performance
- GET /salon/employees/:id/ratings - Get employee ratings

Third-Party Integration Endpoints

Calendar Integration

- POST /integrations/calendar/sync - Sync with external calendar
- DELETE /integrations/calendar/disconnect - Disconnect calendar
- GET /integrations/calendar/status - Get sync status

API Access

POST /integrations/api/token - Generate API token

DELETE /integrations/api/token/:id - Revoke API token

GET /integrations/api/tokens - Get all tokens

System Configuration Endpoints

Language

GET /config/languages - Get available languages

GET /config/translations/:lang - Get translations

Settings

GET /config/settings - Get system settings

PUT /config/settings - Update settings (admin only)

Statistics & Analytics

Public Statistics

GET /stats/salons - Get salon statistics

GET /stats/services - Get popular services

Search & Filter Endpoints

Advanced Search

GET /search/salons - Advanced salon search

GET /search/services - Search services

GET /search/employees - Search employees (within salon)

QR Code Endpoints

QR Management

GET /qr/salon/:salonId - Get salon QR code

POST /qr/scan - Process QR scan

GET /qr/validate/:code - Validate QR code

Reminder & Confirmation

Automated Reminders

- POST /reminders/send - Trigger reminder (system)
- PUT /reminders/:id/confirm - Confirm appointment via reminder
- PUT /reminders/:id/reschedule - Reschedule via reminder link

File Upload Endpoints

Image Upload

- POST /upload/profile-image - Upload profile image
- POST /upload/salon-images - Upload salon images
- DELETE /upload/image/:id - Delete image

Health Check & Utility

- GET /health - Health check
- GET /version - API version

Query Parameters & Filters

Common Query Parameters:

- ?page=1 - Pagination page number
- ?limit=10 - Items per page
- ?sort=createdAt - Sort field
- ?order=desc - Sort order (asc/desc)
- ?search=keyword - Search term
- ?status=active - Filter by status
- ?startDate=2025-01-01 - Filter start date
- ?endDate=2025-12-31 - Filter end date
- ?gender=male - Filter by gender
- ?category=haircut - Filter by category
- ?district=Colombo - Filter by district

?lat=6.9271 - GPS latitude
?lng=79.8612 - GPS longitude
?radius=5 - Search radius (km)

Authentication Headers

Required Headers:

Authorization: Bearer <token> - JWT token
Content-Type: application/json - Request content type
Accept-Language: en - Language preference (en/si)

Standard Response Format

Success Response:

```
{  
  "success": true,  
  "message": "Operation successful",  
  "data": { /* response data */ },  
  "meta": {  
    "page": 1,  
    "limit": 10,  
    "total": 100  
  }  
}
```

Error Response:

```
{  
  "success": false,  
  "message": "Error message",  
  "error": {  
    "code": "ERROR_CODE",  
    "details": []  
  }  
}
```

HTTP Status Codes

200 - OK	- Successful GET, PUT, PATCH
201 - Created	- Successful POST
204 - No Content	- Successful DELETE
400 - Bad Request	- Invalid request data
401 - Unauthorized	- Authentication required
403 - Forbidden	- Insufficient permissions
404 - Not Found	- Resource not found
409 - Conflict	- Resource conflict
422 - Unprocessable Entity	- Validation error
429 - Too Many Requests	- Rate limit exceeded
500 - Internal Server Error	- Server error
503 - Service Unavailable	- Service down

Gaps in the currently developed system

Architecture & Code Organization

Backend Structure Issues:

✗ Current Structure: ✓ Required Structure:

- | | |
|---------------|--------------------|
| - /models | - /src/models |
| - /routes | - /src/controllers |
| - /middleware | - /src/routes |
| - server.js | - /src/services |

Frontend Structure Issues:

✗ Current Structure: ✓ Required Structure:

- | | |
|---------------------------|-----------------------------------|
| - Multiple separate apps | - Monorepo with shared components |
| - No TypeScript | - TypeScript throughout |
| - Basic React | - Modern React with hooks |
| - No internationalization | - i18next for Sinhala/English |

Authentication & Security Gaps

Missing Authentication Features:

- **JWT token system** (no proper token-based auth)
- **OTP verification system**
- **Refresh token mechanism**
- **Password reset functionality**
- **Email verification**
- **Role-based access control middleware**
- **Account lockout after failed attempts**

Security Issues:

- **No input validation middleware**
- **No authentication middleware for protected routes**
- **No request sanitization**
- **No API key management**

Database Design Issues

Incomplete Modles

/ User Model - Missing Fields:

- age, dateOfBirth
- loyaltyPoints
- preferences
- isBlacklisted
- notifications settings

// Salon Model - Missing Fields:

- approvalStatus, description
- amenities, policies
- subscriptionPlan
- socialMedia links
- businessHours (detailed)

// Appointment Model - Missing Fields:

- totalAmount, discountAmount
- paymentStatus, paymentMethod
- reminderSent, confirmationSent
- notes, specialRequests

API Endpoints

Missing Core Endpoints

Authentication

/auth/*/verify-otp

/auth/*/resend-otp

/auth/*/forgot-password

/auth/*/reset-password

/auth/*/refresh-token

/auth/*/logout

Customer Management

/customer/profile

/customer/family-members/*

/customer/preferences

/customer/analytics

/customer/booking-history

Advanced Salon Features

/salon/dashboard

/salon/analytics/*

/salon/employees/*/schedule

/salon/time-slots/block

/salon/reports/*

Admin Features

/admin/dashboard

/admin/analytics/*

/admin/salons/pending

/admin/salons/*/approve

/admin/blacklist/*

/admin/global-services/*

/admin/audit-trail

Payment System

/payments/*

/promotions/*

/loyalty/*

Notification System

- **SMS notifications** (Twilio integration missing)
- **Email notifications** (Nodemailer setup missing)
- **Push notifications**
- **Appointment reminders**
- **Notification preferences**

Payment Integration

- **Payment gateway integration**
- **Payment models and endpoints**
- **Refund management**
- **Invoice generation**

Promotion System

- **Discount codes**
- **Loyalty points system**
- **Promotional campaigns**
- **Offer management**

Analytics & Reporting

- **Dashboard analytics**
- **Revenue tracking**
- **Customer analytics**
- **Service performance metrics**
- **Report generation (CSV/PDF)**

Internationalization (100% missing)

- **Sinhala language support**
- **i18next integration**
- **Language switching**
- **Localized content**

Advanced Search & Filtering (80% missing)

- **GPS-based salon discovery**
- **Advanced filtering options**
- **Search by service type**
- **Availability-based search**

Issues with the Tech Stack

Backend

 Current:	 Required:
- JavaScript	- TypeScript
- Basic Express	- Express + TypeScript
- No validation library	- Zod/Joi validation
- No JWT library	- jsonwebtoken
- No SMS service	- Twilio integration
- No email service	- Nodemailer
- No cron jobs	- node-cron

Frontend

✗ Current:

- JavaScript
- No state management
- No internationalization
- No form validation
- Basic styling
- No API layer

✓ Required:

- TypeScript
- Zustand/React Query
- i18next
- react-hook-form + zod
- TailwindCSS
- Axios with interceptors

Implementation Roadmap

Phase 1: Foundation (Weeks 1-4)

- Migrate to TypeScript
- Restructure code architecture
- Implement proper authentication
- Add security middleware

Phase 2: Core Features (Weeks 5-8)

- Complete API endpoints
- Add notification system
- Implement payment integration
- Enhanced database models

Phase 3: Advanced Features (Weeks 9-12)

- Loyalty and promotion system
- Analytics and reporting
- Internationalization
- Performance optimization